

MECH 632

Intermediate Spaceflight Dynamics

Active Attitude Control (a.k.a. 3-axis stabilization)

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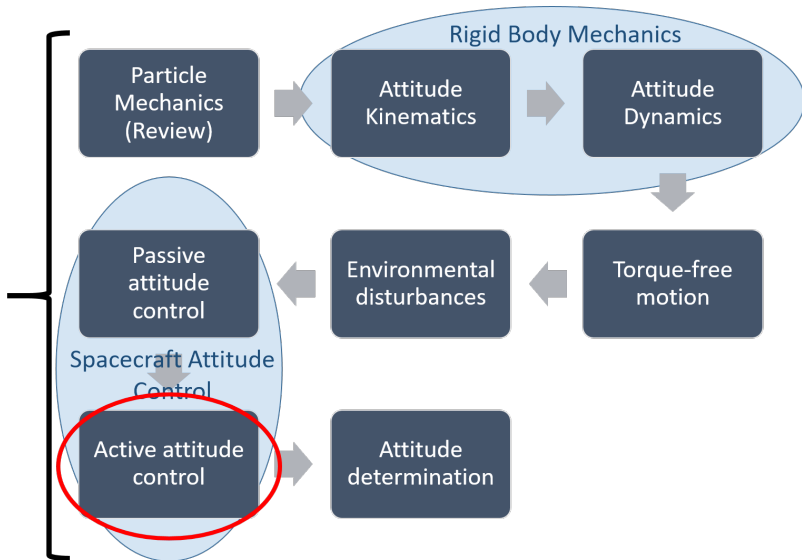
Spring 2019



Where are we?



Spacecraft Attitude Determination and Control





- 1 Lesson Goals
- 2 Spacecraft Attitude Control
 - Linear Quadratic Regulator (LQR)
 - Quaternion-Feedback Control
- 3 Control Actuation with Thrusters
- 4 Control Actuation with Momentum Exchange Devices (MEDs)
- 5 Control Actuation with Magnetic Torquers
- 6 Lesson Summary

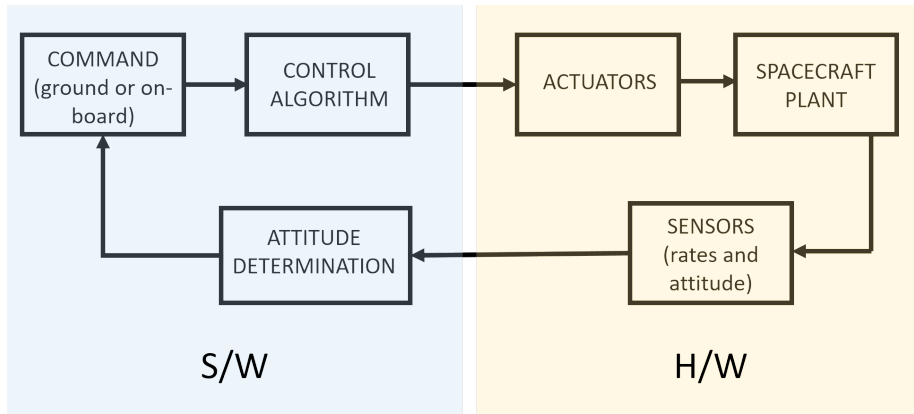


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Lesson Goals



Let's take a closer look at the ADCS block diagram:



Previous Lessons: Modeling the Spacecraft Plant

Lesson Goal: Fill in “Control Algorithm” and “Actuators” blocks.



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- ▶ *Guidance*: defines the desired state; answers the question “Where do I need to go?”
- ▶ *Control*: the action required to follow guidance command, i.e., reduce the error between desired state and current state; answers the question “How do I get there?”
- ▶ *Navigation*: the determination of the current state given sensor measurements; answers the question “Where am I?”
- ▶ *Open-Loop Control*: the system output does not affect the control input
- ▶ *Closed-Loop Control*: the system output affects the control input, i.e., feedback control
- ▶ *Regulation*: Drive the system to a specific state (typically the origin)
- ▶ *Tracking*: Make the system follow a desired trajectory

Linear Systems

SISO:

Root Locus
Bode/Nyquist
Pole placement

MIMO:

State-space methods
Regulator/PID
Optimal Feedback (LQR)

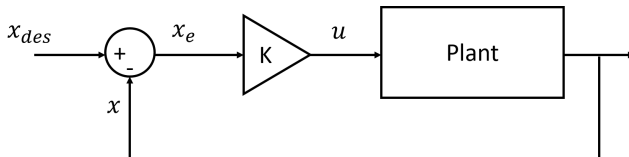
Nonlinear Systems

Feedback Control:

Lyapunov-based
Adaptive Control
Sliding Mode Control

Optimal Control:

Open-loop optimization
Closed-loop optimization



- Assuming the plant is of the form:

$$\dot{x} = Ax + Bu, \quad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m$$

$$y = Cx, \quad C = I_{n \times n}$$

- Controllability:** A linear system is controllable at time t_0 if there exists a time $t_1 > 0$ such that for any $x(t_0)$, there exists an input $u(t)$ that transfers the state to $x(t_1)$. A system is controllable if and only if the following condition is met:

$$\text{rank}(M_C) = \text{rank} \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} = n$$

For $x_{des} = 0$, assume state feedback control of the form:

$$u = -Kx.$$

The closed-loop system is:

$$\dot{x} = (A - BK)x,$$

and is stable if the eigenvalues of $(A - BK)$ have negative real parts.

Selecting gain K :

- ▶ Use pole placement (achieve desired closed-loop poles)
- ▶ Select optimal gain to achieve desired performance...this is LQR

The LQR controller defines the optimal constant K such that:

$$\begin{aligned} \underset{u}{\text{minimize}} \quad & J = \frac{1}{2} \int_0^\infty (x^T Q x + u^T R u) \, dt \\ \text{subject to} \quad & \dot{x} = Ax + Bu. \end{aligned}$$

The system Hamiltonian is: $H = \frac{1}{2} (x^T Q x + u^T R u) + \lambda^T (Ax + Bu)$, where λ is the costate.

Necessary conditions for optimal control:

$$\dot{x} = \frac{\partial H}{\partial \lambda}, \quad \dot{\lambda} = -\frac{\partial H}{\partial x}, \quad \frac{\partial H}{\partial u} = 0.$$

Assuming full state feedback: $u^* = -Kx$, and $\lambda^* = Px$, with K and P constant, the LQR problem is solved by the Algebraic Ricatti Equation (ARE):

$$-Q - A^T P - P A + P B R^{-1} B^T P = 0,$$

and $K = -R^{-1} B^T P$.

- ▶ The ARE is a quadratic matrix equation, and has two solutions (one stable, one anti-stable)...the positive-definite solution is the stable one ($P > 0$).
- ▶ Requirements on user-defined weighting matrices Q and R :
 - ▶ $Q \geq 0$ (positive semi-definite)
 - ▶ $R > 0$ (positive definite, i.e. invertible)
- ▶ Q and R are used as tuning parameters to achieve desired performance

A heuristic approach based on spacecraft capabilities and desired pointing error:

$$Q = \text{diag} \left[\dots, \frac{\alpha_i^2}{x_{max_i}^2}, \dots \right], \text{ for } i = 1, \dots, n$$
$$R = \text{diag} \left[\dots, \frac{\beta_i^2}{u_{max_i}^2}, \dots \right], \text{ for } i = 1, \dots, m$$

where x_{max_i} is the maximum allowable error for the i -th state, u_{max_i} is the maximum allowable value of the i -th control variable, and α_i, β_i are positive constants.

- ▶ Nominally, user selects $\alpha = 1$, $\beta = 1$, and tunes as needed.
- ▶ How do you think changing α and β will affect performance?



- ▶ Advantages of LQR:
 - ▶ Guarantees stability while optimizing performance
 - ▶ Easily implemented (MATLAB command `lqr`)
 - ▶ Linear system allows for independent estimator design (separation principle)
- ▶ Disadvantages of LQR:
 - ▶ Valid only “locally” around linearization point; not an option for large attitude maneuvers
 - ▶ Does not explicitly incorporate constraints (e.g., actuator limits)

Let's look at nonlinear control laws that can be used for large attitude maneuvers.

Recall the quaternion kinematic and dynamic equations:

$$\dot{q} = \frac{1}{2} \begin{bmatrix} [\bar{q}^\times] + q_4 I \\ -\bar{q}^T \end{bmatrix} \omega_{BN}, \quad (1)$$

$$J\dot{\omega}_{BN} = \tau - [\omega_{BN}^\times] J\omega_{BN}. \quad (2)$$

For the rigid spacecraft in circular orbit, those equations become:

$$\dot{q} = \frac{1}{2} \begin{bmatrix} [\bar{q}^\times] + q_4 I \\ -\bar{q}^T \end{bmatrix} \omega_{BO}, \quad (3)$$

$$J\dot{\omega}_{BO} = \tau - [(\omega_{BO} + \omega_{ON})^\times] J(\omega_{BO} + \omega_{ON}), \quad (4)$$

where q is quaternion describing the attitude of B w.r.t. O , and all quantities are resolved in the B CCS.

Assume feedback control of the form:

$$\tau = -K\bar{q}_e + C\omega_e \quad (5)$$

where $\bar{q}_e$ is the vector part of the quaternion error (i.e. the difference between the desired quaternion and the current quaternion), and $\omega_e = \omega_{des} - \omega_{BN}$. The parameters K and C are gain weighting matrices that can be tuned.

Typically $\omega_{des} = \mathbf{0}$ to achieve a stationary pointing condition,

$$\tau = -K\bar{q}_e - C\omega_{BN} \quad (6)$$



From the quaternion successive rotations recall that,

$$q_e = q \otimes q_c^* = G(q_c)^T q$$

where q_c is the commanded (desired) quaternion. The quaternion error is given by:

$$q_e = \begin{bmatrix} q_{4c} & q_{3c} & -q_{2c} & -q_{1c} \\ -q_{3c} & q_{4c} & q_{1c} & -q_{2c} \\ q_{2c} & -q_{1c} & q_{4c} & -q_{3c} \\ q_{1c} & q_{2c} & q_{3c} & q_{4c} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \quad (7)$$

Note: When the desired attitude is the equilibrium point $q_c = [0, 0, 0, 1]^T$, $q_e = q$.



Let's examine the inertial attitude stabilization of a rigid spacecraft to the equilibrium point $q_c = [0, 0, 0, 1]^T$, $\omega_{des} = \mathbf{0}$, and let $\omega = \omega_{BN}$. The quaternion feedback control law is:

$$\tau = -K\bar{q} - C\omega. \quad (8)$$

Define the Lyapunov function:

$$V = \frac{1}{2}\omega^T K^{-1} J \omega + \bar{q}^T \bar{q} + (q_4 - 1)^2.$$

- ▶ Is $V = 0$ at the equilibrium point?
- ▶ Is $V > 0$ at all other points?
- ▶ What is the requirement on K and C for Lyapunov stability?
- ▶ What is the requirement on K and C for local asymptotic stability?



Feedback control law that counteracts the gyroscopic term in Euler's equation:

$$\tau = -K\bar{q}_e - C\omega + [\omega^\times]J\omega \quad (9)$$

To ensure stability we choose $K = kJ$ and $C = cJ$, where k and c are positive scalar constants:

$$\tau = -kJ\bar{q}_e - cJ\omega + [\omega^\times]J\omega \quad (10)$$



- ▶ Optimal Control (open loop); minimum time slew, or minimum control effort slew
- ▶ Robust control
- ▶ Adaptive control
- ▶ Sliding mode control
- ▶ Model Predictive Control (closed-loop optimal control)
- ▶ ...



- ▶ ADCS performance is limited by the hardware used (i.e., actuators, sensors), and design requirements.
- ▶ The *bandwidth* (ω_b) of an attitude control system is an important performance measure.
 - ▶ It is a measure of the “speed” of the response
 - ▶ High bandwidth = short rise time (i.e., “large” in frequency domain is “small” in time domain)
- ▶ Bandwidth of a single-axis ADCS can be approximated by:

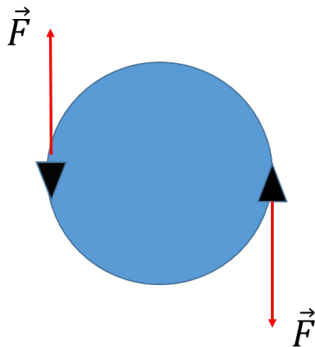
$$\omega_b \approx \sqrt{\frac{\tau_{max}}{\theta_{max}J}},$$

where τ_{max} is the maximum available torque, θ_{max} is the maximum allowable attitude error, and J is the moment of inertia about the axis of interest.



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- ▶ External torque applied by thrusters/reaction jets; a set of two thrusters can create a torque without producing a resultant force
- ▶ Thruster Types:
 - ▶ Cold Gas ($I_{sp} = 30\text{-}120\text{ s}$); nitrogen or argon
 - ▶ Hot Gas ($I_{sp} = 135\text{-}290\text{ s}$); mono-propellant (hydrazine) or bi-propellant
- ▶ Advantages: Simple, can produce large torques, can provide 3-axis maneuverability
- ▶ Disadvantages: Propellant-limited, typically on-off operations (minimum impulse), limited pointing accuracy





- ▶ Thruster-based ADCS use multiple thrusters arranged in a specific manner throughout the body.
 - ▶ Arrangement is not random but carefully designed to meet specific parameters (e.g., torque production, translational force production, plume impingement, etc.).
- ▶ The i -th thruster produces the torque,

$$\boldsymbol{\tau}_i^B = \boldsymbol{r}_i^B \times \boldsymbol{f}_i^B,$$

where $\boldsymbol{r}_i^B$ is the position of the thruster in the body CCS, and $\boldsymbol{f}_i^B$ is the force produced by the thruster in the body CCS.

- ▶ Given N thrusters the torque distribution matrix can be defined as,

$$\boldsymbol{A} = [\boldsymbol{\tau}_1^B \quad \boldsymbol{\tau}_2^B \quad \dots \quad \boldsymbol{\tau}_N^B].$$

- ▶ The Problem: how can we decide which thruster fires and for how long?



- ▶ Thrusters are operated as on/off devices on a given duty cycle (i.e., over a sampling period Δt)
- ▶ Given a requested torque during our sampling period (continuous signal), the thruster mapping problem is posed by,

$$\begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \frac{1}{\Delta t} A \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}$$

- ▶ The underdetermined system has infinitely many solutions if $\text{rank}(A) = 3$.



- ▶ Compute the pseudoinverse of A ,

$$A^+ = \text{pinv}(A).$$

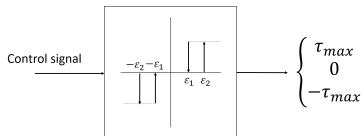
- ▶ Typically the matrix A^+ is chosen such that the norm of the resulting vector is minimized (i.e., Moore-Penrose)
 - ▶ This method can be inefficient for a large number of thrusters.
- ▶ Use a lookup table, i.e. use a predetermined set of thrusters depending on phases of flight, selected modes, and health status.
- ▶ Use a numerical optimization technique to determine the optimal selection of thrusters (see Bong Wie Sec. 7.6.2 for a linear programming approach example).
- ▶ These techniques are implemented within a thrust modulation scheme (essentially a digital-to-analog conversion of the continuous control signal)

The control laws we developed provide a continuous torque signal which has to be modulated onto a discrete signal (i.e., a set of pulses).

Modulation techniques:

- ▶ Schmitt Trigger: simple relay with a deadband and hysteresis
 - ▶ For single-axis control:

$$\text{min. pulse width} = \frac{J(\epsilon_2 - \epsilon_1)}{k_p \tau_{max}}$$



- ▶ More commonly used in spacecraft applications: Pulse-Width Modulation (PWM), and Sigma-Delta Modulation ($\Sigma\Delta M$) (see paper by Zappulla et. al.)

- ▶ The PWM has been widely adopted in the spacecraft community due to its simple implementation.
- ▶ The PWM modulates an analog signal ($v(t)$) onto a series of pulses at a given frequency,

$$f_{PWM} = \frac{1}{T_{PWM}},$$

where T_{PWM} is the PWM sampling time and the analog signal is sampled by a zero-order-hold (ZOH). The output pulse width changes depending on the input signal and maximum signal value (V_{max}). The desired pulsewidth can be computed by:

$$V_{max}\xi = v(t)T_{PWM} \implies \xi = \frac{v(t)}{V_{max}}T_{PWM}$$

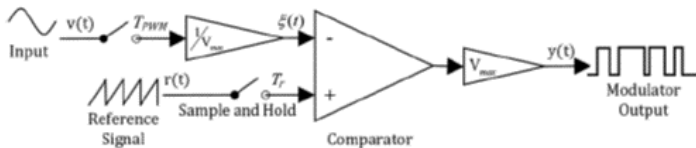
- ▶ In principle: The PWM outputs a series of pulses such that the total angular momentum throughout the PWM cycle is equal to the angular momentum requested by the continuous controller!

PWM Implementation

- ▶ The PWM is implemented using a reference waveform ($r(t)$) which is sampled at a higher frequency:

$$f_r = n f_{PWM}, \quad n = 2, 3, 4, \dots$$

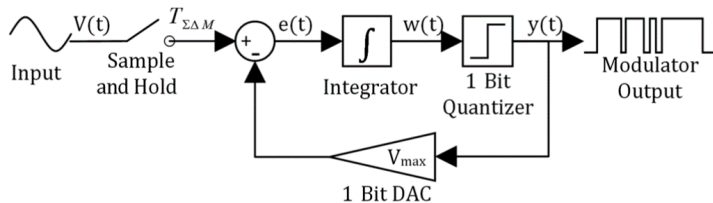
- ▶ The signal $\xi(t)$ is then compared to $r(t)$ and the PWM output is the amount of time for which $\xi(t) \geq r(t)$.



Sigma-Delta Modulator ($\Sigma\Delta M$)



- ▶ Like the PWM, the $\Sigma\Delta M$ modulates an analog signal ($v(t)$) onto a series of pulses, acting like an analog-to-digital converter.
- ▶ The continuous signal is sampled by a ZOH at the modulator frequency $T_{\Sigma\Delta M}$.
- ▶ A negative feedback loop is used to drive the error between the commanded signal and the output signal to zero.
- ▶ The feed-forward path integrates the error signal over the sampling time $T_{\Sigma\Delta M}$, which is modulated by a 1-bit quantizer; pulse triggered when $w(t)$ reaches desired threshold.
- ▶ The feedback path converts the output signal back to analog.





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- ▶ *Momentum Exchange Devices* (MEDs) are rotating wheels inside the spacecraft.
- ▶ Provide *internal* torques that result in an exchange of momentum between the wheel and the spacecraft; total angular momentum is conserved!
- ▶ Provide continuous range of operations (i.e., not on/off devices).
- ▶ Main Types of MEDs:
 - ▶ Momentum Wheel
 - ▶ Reaction Wheel (RW)
 - ▶ Control Moment Gyroscope (CMG) (single-gimbal, double-gimbal, variable-speed)
- ▶ Advantages: high-precision pointing, and disturbance rejection
- ▶ Disdvantages: Require moemntum management (momentum dumping), can have complex steering logic

Let the total angular momentum of the spacecraft be,

$$\mathbf{H}^B = \mathbf{H}_s^B + \mathbf{H}_w^B,$$

where $\mathbf{H}_s^B = J\boldsymbol{\omega}_{BN}$ is the angular momentum of the spacecraft and $\mathbf{H}_w^B$ is the combined angular momentum of the MEDs.

The combined angular momentum of a generic array of N MEDs is written as,

$$\mathbf{H}_w^B = \begin{bmatrix} \hat{a}_1 & \hat{a}_2 & \dots & \hat{a}_N \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_N \end{bmatrix} = Z\mathbf{h},$$

where $\hat{a}_i$ is a unit vector along the direction of the angular momentum for the i -th MED.

Working Principle of MEDs (cont.)

Taking the inertial time derivative of the total angular momentum yields:

$$\dot{\mathbf{H}}^B = \frac{d}{dt} (J\boldsymbol{\omega}_{BN} + Z\mathbf{h}) = \boldsymbol{\tau}^B$$

The general dynamic equation for a spacecraft with MEDs is,

$$J\dot{\boldsymbol{\omega}}_{BN} + [\boldsymbol{\omega}_{BN}^\times]J\boldsymbol{\omega}_{BN} + Z\dot{\mathbf{h}} + \dot{Z}\mathbf{h} + [\boldsymbol{\omega}_{BN}^\times]Z\mathbf{h} = \boldsymbol{\tau}^B. \quad (11)$$

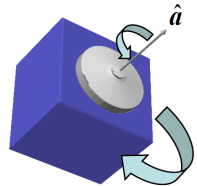
More compactly,

$$J\dot{\boldsymbol{\omega}}_{BN} + [\boldsymbol{\omega}_{BN}^\times]J\boldsymbol{\omega}_{BN} = \boldsymbol{\tau}^B + \boldsymbol{\tau}_w^B \quad (12)$$

$$\boldsymbol{\tau}_w^B = - \left(Z\dot{\mathbf{h}} + \dot{Z}\mathbf{h} + [\boldsymbol{\omega}_{BN}^\times]Z\mathbf{h} \right) \quad (13)$$

Reaction Wheels (RWs)

- ▶ A rotor fixed to the spacecraft rotating at a variable speed.
- ▶ Motor provides torque to the wheel in order to accelerate/decelerate it (typically brushless motors are used).



- ▶ Applying power to the motor produces a torque on the wheel, thus increasing its angular momentum; this causes an equal and opposite torque and angular momentum on the spacecraft (total angular momentum is conserved).
- ▶ Angular momentum of the RW: $\vec{h} = J_w w \hat{a}$
 - J_w : RW inertia about its spin axis
 - w : RW speed
 - $\hat{a}$: unit vector along RW angular momentum



- ▶ RWs are often used to counteract disturbances affecting the spacecraft's attitude; how can we know how big of a RW wheel to use? i.e., How much angular momentum must the RW be able to store in order to counteract the disturbance?
- ▶ Consider a single-axis example, with an external disturbance torque, and a RW mounted along this axis. The EOM yields:

$$\dot{H} = \dot{H}_s + \dot{H}_w = \tau_{dist}$$

- ▶ Since the RW will exactly counteract the disturbance, the spacecraft will remain stationary, and we can say that,

$$\begin{aligned} \dot{H}_w &= \tau_{dist} \\ H_w &= \int_{t_0}^{t_f} \tau_{dist} dt \end{aligned} \quad (14)$$

- ▶ H_w is the total angular momentum required to exactly counteract the disturbance.

The angular momentum contribution of a RW cluster is:

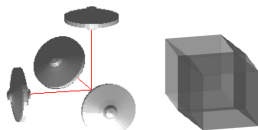
$$\mathbf{h}^B = \sum_i^N \mathbf{h}_i = [\hat{a}_1 \quad \hat{a}_2 \quad \dots \quad \hat{a}_N] \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_N \end{bmatrix} = Z \begin{bmatrix} J_{w1}w_1 \\ J_{w2}w_2 \\ \vdots \\ J_{wN}w_N \end{bmatrix}$$

Since the $\hat{a}_i$ unit vectors are fixed to the body, $\dot{Z} = 0$ and Eq. 13 becomes,

$$\boldsymbol{\tau}_w^B = - \left(Z \dot{\mathbf{h}} + [\boldsymbol{\omega}_{BN}^\times] Z \mathbf{h} \right) = - \left(Z \begin{bmatrix} J_{w1}\dot{w}_1 \\ J_{w2}\dot{w}_2 \\ \vdots \\ J_{wN}\dot{w}_N \end{bmatrix} + [\boldsymbol{\omega}_{BN}^\times] Z \begin{bmatrix} J_{w1}w_1 \\ J_{w2}w_2 \\ \vdots \\ J_{wN}w_N \end{bmatrix} \right)$$

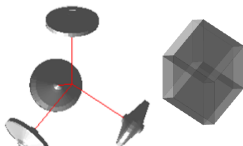
Note: $\dot{\mathbf{h}}$ is the term we wish to control in order to produce the desired torque.

- ▶ How many RWs are required to control motion about all 3 principal axes? Should redundancy be considered?
- ▶ Typical RW arrays:



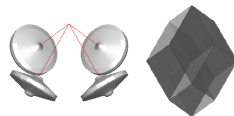
NASA's "Ortho-Skew" Array

(a)



Loral's Array

(b)



Regular Pyramid

(c)

- ▶ Exercise: Write the Z matrix for these RW arrays.
- ▶ How should we deal with failed RWs?

- ▶ Goal: Compute the control input ($\dot{\mathbf{h}}$, i.e. $\dot{w}_i$) to the RWs given a required torque (from controller)

$$\boldsymbol{\tau}_w^B = - \left(Z\dot{\mathbf{h}} + [\boldsymbol{\omega}_{BN}^\times] Z\mathbf{h} \right) = \boldsymbol{\tau}_{req}$$

$$\boxed{\dot{\mathbf{h}} = Z^{-1} \left(-\boldsymbol{\tau}_{req} - [\boldsymbol{\omega}_{BN}^\times] Z\mathbf{h} \right)} \quad (15)$$

- ▶ $\dot{\mathbf{h}}$ is the control command to the wheels (i.e., the torque applied to the wheels in order to change their speed).
- ▶ Steering Law requires to compute Z^{-1} ; what is the dimension of Z ?
- ▶ Typically $\text{rank}(Z) = 3$, and the pseudo-inverse, $Z^+ = \text{pinv}(Z)$ is used:

$$\boxed{\dot{\mathbf{h}} = Z^+ \left(-\boldsymbol{\tau}_{req} - [\boldsymbol{\omega}_{BN}^\times] Z\mathbf{h} \right)} \quad (16)$$



- ▶ RWs have limitations based on their design; the commanded torque $\dot{\mathbf{h}}$ may not be able to be realized by the RW.
- ▶ Saturation Logic:

$$\begin{aligned}\tau_{w_{min}} &\leq \dot{h}_i \leq \tau_{w_{max}}, \text{ for } i = 1, \dots, N \\ h_{w_{min}} &\leq \mathbf{h}_i \leq h_{w_{max}}, \text{ for } i = 1, \dots, N\end{aligned}$$

- ▶ With the saturation logic implemented, the effective torque applied to the spacecraft can be computed:

$$\tau_{RW} = - \left(Z\dot{\mathbf{h}} + [\boldsymbol{\omega}_{BN}^{\times}]Z\mathbf{h} \right) \quad (17)$$

- ▶ The effective torque τ_{RW} is the control input into the plant (spacecraft) (detailed block diagram on the board...).

Manufacturer	Ang. Momentum	Torque	Power
Honeywell	4 – 250 Nms	0.12 – 1.2 Nm	--
Blue Canyon Tech	1 – 8 Nms	0.1 – 0.25 Nm	4.5 – 10 W
Blue Canyon Tech	0.015 – 0.5 Nms	0.004 – 0.025 Nm	0.6 – 6 W
Sinclair	0.06 Nms	0.005 Nm	0.2 – 7 W

Pros:

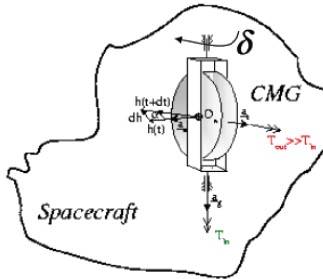
- ▶ Reliable and high pointing accuracy
- ▶ No propellant consumption
- ▶ Simple steering law

Cons:

- ▶ Cannot produce large torque (beyond $1Nm$)
- ▶ Requires momentum management/momentum dumping strategy

Control Moment Gyroscopes (CMGs)

- ▶ A CMG is a spinning wheel that gimbals about an axis fixed on the spacecraft body.
- ▶ Input torque along the gimbal axis changes the direction of the wheel's angular momentum vector; this produces a torque on the spacecraft in the opposite direction.



$$||\tau_{in}|| \approx J_g \ddot{\delta}$$

$$||\tau_{out}|| \approx h_w \dot{\delta} = J_w w \dot{\delta}$$

$$||\tau_{out}|| \gg ||\tau_{in}|| \text{ since } w \gg \ddot{\delta}$$

Three main types:

- ▶ Single-Gimbal CMG (SG-CMG)
- ▶ Double-Gimbal CMG (DG-CMG)
- ▶ Variable-Speed CMG (VS-CMG)

The angular momentum contribution of a SG-CMG cluster is:

$$\mathbf{h}^B = \sum_i^N \mathbf{h}_i = [\hat{a}_1(\delta_1) \quad \hat{a}_2(\delta_1) \quad \dots \quad \hat{a}_N(\delta_N)] \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_N \end{bmatrix} = Z(\boldsymbol{\delta}) \begin{bmatrix} J_{w1}w_1 \\ J_{w2}w_2 \\ \vdots \\ J_{wN}w_N \end{bmatrix}$$

Since the CMG wheel is spinning at a constant rate $\dot{\mathbf{h}} = 0$ and Eq. 13 becomes,

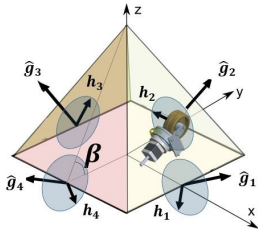
$$\boldsymbol{\tau}_w^B = - \left(\dot{Z}(\boldsymbol{\delta})\mathbf{h} + [\boldsymbol{\omega}_{BN}^\times]Z(\boldsymbol{\delta})\mathbf{h} \right), \text{ with } \dot{Z}(\boldsymbol{\delta}) = \frac{\partial Z}{\partial \boldsymbol{\delta}}\dot{\boldsymbol{\delta}} = A(\boldsymbol{\delta})\dot{\boldsymbol{\delta}}$$

Assuming all CMGs in the array have the same angular momentum magnitude:

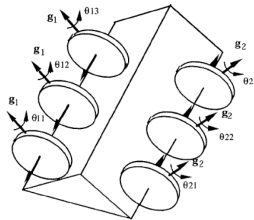
$$\boldsymbol{\tau}_w^B = - \left(A(\boldsymbol{\delta})\dot{\boldsymbol{\delta}}J_w w + [\boldsymbol{\omega}_{BN}^\times]Z(\boldsymbol{\delta})\mathbf{h} \right)$$

SG-CMG Clusters (cont.)

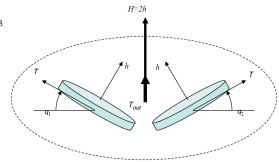
- ▶ AS with RWs, 3 CMGs are required to control motion about all 3 principal axes.
- ▶ Typical CMG arrays:



(a) Pyramid Array



(b) Roof Array



(c) Scissored Pair

- ▶ Pyramid array is the most well-studied; Exercise: write the Z and A matrices for a generic 4-CMG pyramid array with skew-angle β .

- ▶ Goal: Compute the control input ($\dot{\delta}$) to the CMGs given a required torque (from controller)

$$\tau_w^B = - \left(A(\delta) \dot{\delta} J_w w + [\omega_{BN}^\times] Z(\delta) h \right) = \tau_{req}$$

$$\boxed{A(\delta) \dot{\delta} = \frac{1}{J_w w} \left(-\tau_{req} - [\omega_{BN}^\times] Z(\delta) h \right)} \quad (18)$$

- ▶ The matrix $A(\delta)$ must be inverted; as we did with the RW steering law, the pseudoinverse can be used:

$$A^+ = \text{pinv}(A) = A^T (A A^T)^{-1} \quad (19)$$

- ▶ *Singularities:*

- ▶ If $\text{rank}(A) < 3$, or equivalently $\text{rank}(A A^T) < 3$, for a set of gimbal angles the pseudoinverse does not exist (i.e., the CMG steering logic encounters a singular state and $\dot{\delta}$ goes to infinity.
- ▶ The singularity condition is (*singularity measure*),

$$\det(A A^T) = 0$$



- ▶ Null Motion:

$$\dot{\delta} = A^T(AA^T)^{-1}\dot{\mathbf{h}} + \gamma\mathbf{n},$$

where γ is a scalar representing the amount of null motion to be added, and $\mathbf{n}$ is a null vector of A .

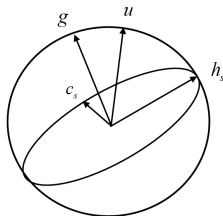
- ▶ Singularity-robust inverse:

$$A^\# = A^T(AA^T + \lambda I)^{-1},$$

where λ is a small positive constant of the order of 0.01.

These pseudoinverse-based approaches do not *guarantee* singularity avoidance and often overconstrain the operational momentum envelope of the SG-CMG system. Other “global” approaches tend to be computationally expensive, but include:

- ▶ Constrained optimal control (i.e., find optimal gimbal angles that maximize singularity measure).
- ▶ Ground-based feed-forward path planning.



g : Gimbal axis unit vector
 u : Unit singular vector
 h_s : Singular momentum vector for given u
 c_s : Actual unit torque vector from h_s

- ▶ Singularity occurs when for any singular vector $\hat{u}$ the torque vector $\hat{c}_s$ is perpendicular to $\hat{u}$ (corresponding momentum vector is either h_s or $-h_s$).
- ▶ For the i -th CMG the unit torque vector in the singular condition is,

$$\hat{c}_{si} = \epsilon_i \frac{\hat{g}_i \times \hat{u}}{\|\hat{g}_i \times \hat{u}\|}, \text{ with } \epsilon_i = \pm 1.$$

- ▶ The total singular angular momentum surface for the N -CMG array is,

$$\vec{H}_s = \sum_{i=1}^N (\hat{c}_{si} \times \hat{g}_i).$$



- ▶ As with RWs, CMGs have physical limitations that must be accounted for.
- ▶ Saturation Logic:

$$\dot{\delta}_{min} \leq \dot{\delta}_i \leq \dot{\delta}_{max}, \text{ for } i = 1, \dots, N$$

- ▶ Singularity measure:

$$m = \sqrt{AA^T}$$

- ▶ With the saturation logic implemented, the effective torque applied to the spacecraft can be computed:

$$\tau_{CMG} = - \left(A(\delta) \dot{\delta} J_w w + [\omega_{BN}^\times] Z(\delta) h \right) \quad (20)$$

- ▶ The effective torque τ_{RW} is the control input into the plant (spacecraft) (detailed block diagram on the board...).

Manufacturer	Ang. Momentum	Torque	Power
L3	4760 Nms	258 Nm	--
Airbus/Collins	15 Nms	45 Nm	25 W

Pros:

- ▶ High pointing accuracy with no propellant use
- ▶ Much higher torque-to-power ration than RWs
- ▶ Can provide torque for agile maneuvering; torque levels not possible with RWs

Cons:

- ▶ More costly and generally less reliable than RWs due to complex design
- ▶ Complex steering logic with singularities



- ▶ In some cases using VS-CMG, or a mixtures of CMGs and RWs can help mitigate the issues of singularities.
- ▶ Let's look at a mixed array of RWs and VS-CMGs:

$$\mathbf{H}^B = \mathbf{H}_s^B + \mathbf{H}_w^B = \boldsymbol{\tau}^B, \text{ with } \mathbf{H}_w^B = \mathbf{h}_{RW} + \mathbf{h}_{CMG}$$

- ▶ Define the vector of controlled quantities, $\mathbf{c} = [\mathbf{w}_{RW}, \mathbf{w}_{CMG}, \boldsymbol{\delta}]^T$, then

$$\begin{aligned} \frac{N d}{dt} \mathbf{H}_w^B &= \frac{B d}{dt} \mathbf{H}_w^B + [\boldsymbol{\omega}_{BN}^\times] \mathbf{H}_w^B \\ \frac{B d}{dt} \mathbf{H}_w^B &= \frac{\partial \mathbf{H}_w^B}{\partial \mathbf{c}} \dot{\mathbf{c}} = \mathbf{A} \mathbf{u} \end{aligned}$$

- ▶ Total EOM:

$$\mathbf{J} \dot{\boldsymbol{\omega}}_{BN} + [\boldsymbol{\omega}_{BN}^\times] \mathbf{J} \boldsymbol{\omega}_{BN} = \boldsymbol{\tau}^B + \boldsymbol{\tau}_w^B$$

$$\boxed{\boldsymbol{\tau}_w^B = -(\mathbf{A} \mathbf{u} + [\boldsymbol{\omega}_{BN}^\times] \mathbf{H}_w^B)}$$



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Magnetorquers (MTs)

- ▶ Conductive wire wrapped around a ferromagnetic core (a.k.a. torque rod); applying current to the coil generates a magnetic dipole which interacts with the ambient magnetic field to produce an external torque:

$$\vec{\tau}_{MT} = \vec{m} \times \vec{B},$$

where $\vec{m}$ is the magnetic dipole moment and $\vec{B}$ is the ambient magnetic field.

- ▶ Generated torque is always perpendicular to the magnetic field.
- ▶ Working principle: $\vec{B}$ is known based on a given on-board model (usually known in ECEF and must be transformed to the Body CCS); attitude control law provides a requested torque,

$$\tau_{MT}^B = [\mathbf{m}^\times] \mathbf{B}$$

Goal: find the appropriate $\mathbf{m}$ to produce the requested torque.



- ▶ The torque expression cannot be simply inverted to “solve” for $\mathbf{m}$.
- ▶ Let's cross both sides of the equation with $\vec{B}$:

$$\begin{aligned}\vec{B} \times \vec{\tau}_{MT} &= \vec{B} \times (\vec{m} \times \vec{B}) \\ \vec{B} \times \vec{\tau}_{MT} &= (\vec{B} \cdot \vec{B})\vec{m} - (\vec{m} \cdot \vec{B})\vec{B}\end{aligned}$$

- ▶ Choose $\vec{m}$ such that not component exists along $\vec{B}$ which would not generate any torque (i.e., pick $\vec{m} \perp \vec{B} \implies \vec{m} \cdot \vec{B} = 0$).
- ▶ The resulting steering law is:

$$\mathbf{m} = \frac{[\mathbf{B}^\times] \boldsymbol{\tau}_{MT}}{\mathbf{B}^T \mathbf{B}}$$

- ▶ Since MTs provide an external torque they are often used in conjunction with MEDs to provide momentum dumping capability.
- ▶ As the MEDs gradually build up angular momentum, MTs can provide an external torque causing the MEDs to react and decrease their angular momentum.
- ▶ In this case, the requested torque for the MTs is,

$$\tau_{MT} = -k\mathbf{H}_w^B,$$

where k is a positive constant gain, and $\mathbf{H}_w^B$ is the angular momentum of the MED array.

- ▶ The same steering law can be used to produce the momentum dumping torque.



- ▶ Detumbling control is another popular use of MTs; many CubeSats exit the launch vehicle dispenser in a tumbling state, and require detumbling before ADSC functions can operate properly.
- ▶ MTs provide an effective (yet slow) way of detumbling a spacecraft as it travels along its orbital path.
- ▶ “B-dot” Steering Law:

$$\mathbf{m} = -k\dot{\mathbf{B}}, \quad k > 0$$



- ▶ As with other actuators, MTs have physical limitations:

$$m_{min} \leq m_i \leq m_{max}, \text{ for } i = 1, \dots, N$$

- ▶ With the saturation logic implemented, the effective torque applied to the spacecraft can be computed:

$$\tau_{MT} = [\mathbf{m}^\times] \mathbf{B} \quad (21)$$

- ▶ The effective torque τ_{MT} is the control input into the plant (spacecraft)



Pros:

- ▶ Reliable and inexpensive
- ▶ No propellant use

Cons:

- ▶ Small torque output (about 10^{-3} Nm)
- ▶ Limited pointing capability
- ▶ Strongly dependent on orbit



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- ▶ We looked at two different feedback controllers that can be used for 3-axis attitude control (LQR, quaternion feedback).
- ▶ Torque requested by the control algorithm is realized by actuators:
 - ▶ Thrusters
 - ▶ Reaction Wheels
 - ▶ Control Moment Gyroscopes
 - ▶ Magnetorquers
- ▶ We looked at pros/cons of each actuator type and defined steering laws for each.